

MiniSpeedCam r1.1 — PCB Design Review

Valar Systems — Hardware Engineering · Doc MSC-R1.1-DR-001 Rev A · 2026-06-22 · **CONFIDENTIAL**

Radar speed-camera motherboard (ESP32-S3 + STM32F301 + CDM324 24 GHz Doppler) · 4-layer, 50 × 50 mm
Repo: [Valar-Systems/MiniSpeedCam @ version/r1.1](#) · Method: kicad-happy analyzer suite

Components	Stack	Routed	EMC risk	Errors / Warnings
94	4-layer (F / GND / PWR / B), 1.6 mm	100%	10 / 100	7 / 32

1. Scope & Method

Produced by cloning the open-source [kicad-happy](#) review suite and running its Python analyzers against the **r1.1** KiCad source ([version/r1.1/hardware/pcb/kicad/](#)). Findings carry analyzer rule IDs and were triaged by hand against the raw netlist.

Analyzer	Status	Result
analyze_schematic.py	RAN	94 parts, 101 nets, 37 unique — 1 error / 3 warnings
analyze_pcb.py --full	RAN	4-layer, 50×50 mm, fully routed — 6 errors / 26 warnings
analyze_gerbers.py	RAN	11 layers complete, PTH+NPTH — 1 warning
cross_analysis.py	RAN	6 cross-domain findings
analyze_emc.py (44 rules)	RAN	65 findings, EMC risk 10/100 (conservative)
Datasheet verification	PARTIAL	Front-end supply/ground pins confirmed (\$6); op-amp signal-pin order is a designer check
analyze_thermal.py	PARTIAL	Ran, but no power-budget input → junction temps not computed
SPICE	SKIPPED	No simulator on host

Layer stack: [F.Cu](#) / [In1.Cu = GND plane](#) / [In2.Cu = split power plane](#) / [B.Cu](#) , 1.6 mm.

2. Executive Summary

The board is complete, fully routed, and mechanically tidy. The issues that matter cluster around one theme: **the sensitive 24 GHz Doppler analog front-end is not adequately isolated from the digital/RF noise of the ESP32-S3**. This is the electrical root cause behind the firmware workarounds in the codebase (a radar-blanking handshake that discards FFT frames during every WiFi burst, plus WiFi TX power reduced to 11 dBm). A final spin should spend its budget here. The rest are standard pre-fab DFM hygiene items.

Top priority — Give the radar amplifier chain (U4 LMP7731, U12 MAX9814) a dedicated low-noise analog supply and a single-point analog ground, instead of the shared digital `3V3` / `GND`. This is the change most likely to remove the need for firmware radar-blanking and let WiFi TX power return to full range.

3. Priority Finding — Radar Front-End Shares the Digital Supply & Ground

```
RADAR SUPPLY [GOOD: dedicated, filtered diode rail]
USB 5V (VUSB)
  -> U5 B0505S (isolated DC-DC) -> +Vo
    -> FB1 ferrite -> CDM324 V+

U5.1 (-Vin) = GND    and    U5.3 (0V, secondary gnd) = GND
[isolation collapsed to a common ground]

RADAR SIGNAL [uV signal amplified on the noisy digital rail]
CDM324 IF out -> C11/R12 (AC couple)
  -> U4 LMP7731 (gain)    [U4.V+  = 3V3]
  -> U12 MAX9814 (AGC)    [U12.VDD = 3V3]
  -> ADC
```

Observation	Evidence (netlist)	Severity
Doppler amplifiers on shared digital rail	<code>U4.5 (V+) -> 3V3</code> and <code>U12.5 (VDD) -> 3V3</code> — same rail as ESP32-S3/STM32 logic; a WiFi-TX transient on 3V3 lands in the gain stage. Both supply pins datasheet-confirmed (§6).	HIGH
Isolated converter ground bonded to digital ground	<code>U5.1 (-Vin) -> GND</code> and <code>U5.3 (0V) -> GND</code> ; no analog/isolated ground net exists in the design. The B0505S isolation is unused — radar return shares digital ground.	HIGH
Diode rail done well	<code>U5.4 (+Vo) -> FB1 -> J3.3 (CDM324 V+)</code> with C22 bulk — a separate, ferrite-filtered radar supply.	OK

Recommended changes

- Derive a clean analog rail for U4/U12 from the isolated B0505S `+Vo` via a small **low-noise (high-PSRR) LDO**, not the digital `3V3`.
- Give the front-end a local **analog ground island**, tied to digital `GND` at a **single star point** near the STM32 ADC return.
- If the B0505S stays, decide deliberately whether its secondary `0V` joins `GND` at one point (quiet) or not at all (true isolation). Today it's a common-ground bond, giving up most of the part's benefit.

Note: `VOUT` is the MAX9814 audio output to the ADC, not a supply — so `RS-001 "VOUT has no source"` is a false positive.

4. Ground & Reference-Plane Integrity

Rule	Severity	Finding & action
PS-002 / GP-001	ERROR	<code>GND</code> resolves into 6 copper islands, 2 signals crossing a gap ; <code>3V3</code> into 23 islands , 2 crossing. Verify <code>In1.Cu</code> is a continuous ground reference (no traces carved through it) and stitch the pours. 3V3 fragmentation matters here because the analog amps reference 3V3.
RP-001	ERROR	Camera <code>PCLK</code> changes layers with no return-path stitching via ; same for several STM/UART/BOOT0 nets. Add a ground via at each clock layer-transition.

5. EMC Pre-Compliance & DFM

Rule	Severity	Finding & action
IO-001	ERROR	No filtering on J1 (Phoenix power input), USB1, USB2 . Add TVS + pi/ferrite on the J1 inlet; common-mode choke + ESD on each USB-C data pair.
CK-001 / CK-003	WARN	Camera <code>XCLK</code> / <code>PCLK</code> and <code>SD_HOST_CLK</code> on outer layers, routed near connectors J8/J10 — move inner-layer or add guard ground.
DC-003	WARN	13 decoupling caps (C6/C7/C9/C12/C17/C42...) far from their plane via — shorten cap→via path to cut PDN inductance.
FD-001	ERROR	No fiducials either side (fine-pitch QFN present) — add ≥ 3 global fiducials.
PM-002	ERROR	C6 and FB1 sit 0.25 mm from board edge — move in ≥ 0.5 mm (crack risk on depanel). J1/J3 overhang is intentional edge-mount — ignore.
TE-001	WARN	0 test points . Add TPs on every rail, the radar analog output, and GND scope points.
DFM-001	WARN	0.1 mm / 0.116 mm trace+space forces JLC advanced tier. Relax non-camera nets to 0.127 mm to drop to standard if cost matters.
TV-001	WARN	STM32 (U13) thermal pad has 5/9 recommended vias — top up to 9 (minor).
SS-001	ERROR	MPN coverage 3/37 in symbols. Populate MPN fields from the existing <code>MiniSpeedCam_BOM.csv</code> LCSC column — BOM hygiene, not an electrical defect.

6. Datasheet Verification

Part	Pin checked	Result	Basis
U5 B0505S-1W	1 = -Vin, 2 = +Vin, 3 = -Vout(0V), 4 = +Vout	CONFIRMED	Standard B0505S 4-pin isolated DC-DC pinout. -Vin and -Vout both on GND verified.
U12 MAX9814	5=VDD, 6=MICOUT, 8=MICIN, GND=4/7/11/EP	CONFIRMED	Exact match to documented MAX9814 14-pin TDFN. VDD → 3V3 verified.
U4 LMP7731	5 = V+	CONFIRMED	Pin 5 = V+ for SOT-23-5 single op-amp. V+ → 3V3 verified.
U4 LMP7731	1-4 = OUT/+IN/-IN order	DESIGNER CHECK	Symbol uses standard SOT-23-5 order (1=OUT, 2=V-, 3=+IN, 4=-IN). Datasheet figure couldn't be auto-rendered here — confirm visually against the LMP7731 connection diagram before fab. Not a flagged discrepancy.

7. Triaged as False-Positive / Expected (no action)

- [CC-002](#) (16×) "narrow signal" on camera/I²C nets — 0.1 mm is fine for low-current signals.
- [GR-004](#) B-paste 26/187 apertures — back side has only 2 components; expected.
- [DP-005](#) USB diff-pair 7% / 1.3 mm mismatch — negligible at USB full-speed.
- [PM-001](#) courtyard overlaps (U15/C2, U16/C26, ...) — decoupling caps deliberately under their LDOs.
- [RS-001](#) on [VOUT](#) — [VOUT](#) is a signal, not a power rail.

8. Review Gaps & Confidence

- No full datasheet sync.** Front-end supply/ground pins spot-verified (§6); LMP7731 I/O pin order and STM32/ESP pin maps not exhaustively checked against rendered figures.
- No SPICE simulation** (no simulator on host): gain/AGC/filter cutoffs not numerically verified.
- Thermal inconclusive:** no power-budget data, so junction temps not computed. Unlikely to be a blocker, but not positively cleared.

Bottom line: r1.1 is electrically functional and routable, but its radar receive path is coupled to the digital domain through both supply (3V3) and ground. Fixing that — a dedicated low-noise analog rail + single-point analog ground for U4/U12, a continuous ground reference, and basic I/O filtering — is the highest-value content for a final spin and directly targets the noise problem the firmware is currently working around.

Generated with the kicad-happy analyzer suite. Rule IDs reference kicad-happy detectors. Findings are engineering guidance for human review, not a substitute for DRC/ERC sign-off or full datasheet verification.